Hints: Integrals – 3. Differentiation under the integral sign and Frullani integrals

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Work with $\int_\varepsilon^R$ and split it into two integrals; in each substitute $t = ax$ resp. $t = bx$. Everything cancels except $\int_{a\varepsilon}^{b\varepsilon} \frac{f(t)}{t}\,dt - \int_{aR}^{bR} \frac{f(t)}{t}\,dt$.

Hint 2. By the mean value theorem for integrals, $\int_{a\varepsilon}^{b\varepsilon} \frac{f(t)}{t}dt = f(\xi)\ln\frac{b}{a} \to f(0)\ln\frac{b}{a}$, and similarly the second piece tends to $f(\infty)\ln\frac{b}{a}$. Hence $\int_0^\infty \frac{e^{-ax}-e^{-bx}}{x}dx = \ln\frac{b}{a}$. For cos, $f(\infty)$ does not exist, but $\int_{aR}^{bR} \frac{\cos t}{t}dt \to 0$ (integrate by parts once), so the same proof gives $\ln\frac{b}{a}$.

Hint 3. For $F(b) = \int_0^1 \frac{x^b - x^a}{\ln x}dx$: $F'(b) = \int_0^1 x^b\,dx = \frac{1}{b+1}$ and $F(a) = 0$, so $F(b) = \ln\frac{b+1}{a+1}$ (for $a, b > -1$).

Problem 2

Note. the differential equation only gives $I(\alpha) = I(0)e^{-\alpha^2/4}$; the value $I(0) = \frac{\sqrt{\pi}}{2}$ is not a "special case" of it and needs a separate argument (Hint 3).

Hint 1. Differentiate under the integral sign (justified: $|xe^{-x^2}\sin\alpha x| \leqslant xe^{-x^2}$ is integrable, uniformly in α): $I'(\alpha) = -\int_0^\infty xe^{-x^2}\sin(\alpha x)\,dx$. Integrate by parts with $dv = xe^{-x^2}dx$, $v = -\frac{1}{2}e^{-x^2}$.

Hint 2. You get $I' = -\frac{\alpha}{2}I$, so $I(\alpha) = I(0)e^{-\alpha^2/4}$.

Hint 3. For $I(0)$ use another parameter: $G(t) = \left(\int_0^t e^{-x^2}dx\right)^2 + \int_0^1 \frac{e^{-t^2(1+u^2)}}{1+u^2}du$. Show $G'(t) = 0$ (substitute $x = tu$ in the first term's derivative), so $G(t) = G(0) = \frac{\pi}{4}$; letting $t \to \infty$ gives $\left(\int_0^\infty e^{-x^2}\right)^2 = \frac{\pi}{4}$.

Problem 3

Hint 1. $I'(a) = \int_0^\pi \frac{2a - 2\cos x}{1 - 2a\cos x + a^2}dx$ for $|a| \neq 1$. Write $2a - 2\cos x = \frac{1}{a}\left[(1 - 2a\cos x + a^2) - (1 - a^2)\right]$ and use $\int_0^\pi \frac{dx}{1 - 2a\cos x + a^2} = \frac{\pi}{|1-a^2|}$ (half-angle substitution).

Hint 2. So $I'(a) = 0$ for $|a| < 1$ and $I'(a) = \frac{2\pi}{a}$ for $|a| > 1$. Since $I(0) = 0$, $I \equiv 0$ on $(-1, 1)$. For $|a| > 1$ factor out a^2: $I(a) = 2\pi\ln|a| + I(1/a) = 2\pi\ln|a|$.

Hint 3. For $a = \pm 1$: $\int_0^\pi \ln(2 \mp 2\cos x)\,dx = \int_0^\pi \ln\left(4\sin^2\frac{x}{2}\right)dx$ (or with cos), which is 0 by $\int_0^{\pi/2} \ln\sin u\,du = -\frac{\pi}{2}\ln 2$. Answer: 0 for $|a| \leqslant 1$, $2\pi\ln|a|$ for $|a| > 1$.

Problem 4

Hint 1. Let $F(\alpha) = \int_0^\infty e^{-\alpha x}\frac{\sin x}{x}dx$, $\alpha \geqslant 0$. For $\alpha \geqslant \alpha_0 > 0$ the derivative integral $-\int_0^\infty e^{-\alpha x}\sin x\,dx$ converges uniformly, and equals $-\frac{1}{1+\alpha^2}$.

Hint 2. So $F(\alpha) = C - \arctan\alpha$; since $|F(\alpha)| \leqslant \int_0^\infty e^{-\alpha x}dx = \frac{1}{\alpha} \to 0$, $C = \frac{\pi}{2}$.

Hint 3. The delicate part is continuity of F at 0^+. Show the tails are uniformly small: for $\alpha \geqslant 0$, $\left|\int_A^\infty e^{-\alpha x}\frac{\sin x}{x}dx\right| \leqslant \frac{2}{A}$ (integrate by parts, or use the second mean value theorem, since

$e^{-\alpha x}/x$ decreases in x). Then F is a uniform limit of continuous functions on $[0,\infty)$, and $F(0) = \lim_{\alpha\to 0^+} F(\alpha) = \frac{\pi}{2}$.

Problem 5

Hint 1. Use $\int_{-\infty}^{\infty} e^{-cx^2}dx = \sqrt{\pi/c}$. (a) complete the square: $-2x^2 - 5x - 3 = -2(x + \frac{5}{4})^2 + \frac{1}{8}$. (b) shift $x \mapsto x + 2024$. (c) substitute $u = x^2$ resp. $u = x^4$ to get Gamma integrals. (d) substitute $u = 1/t$.

Hint 2. Answers: (a) $e^{1/8}\sqrt{\pi/2}$; (b) $2\sqrt{\pi}$; (c) $\frac{1}{2}\Gamma(2) = \frac{1}{2}$ and $\frac{1}{4}\Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{8}$; (d) $\int_0^{\infty} e^{-2019u^2/4}du = \sqrt{\pi/2019}$.